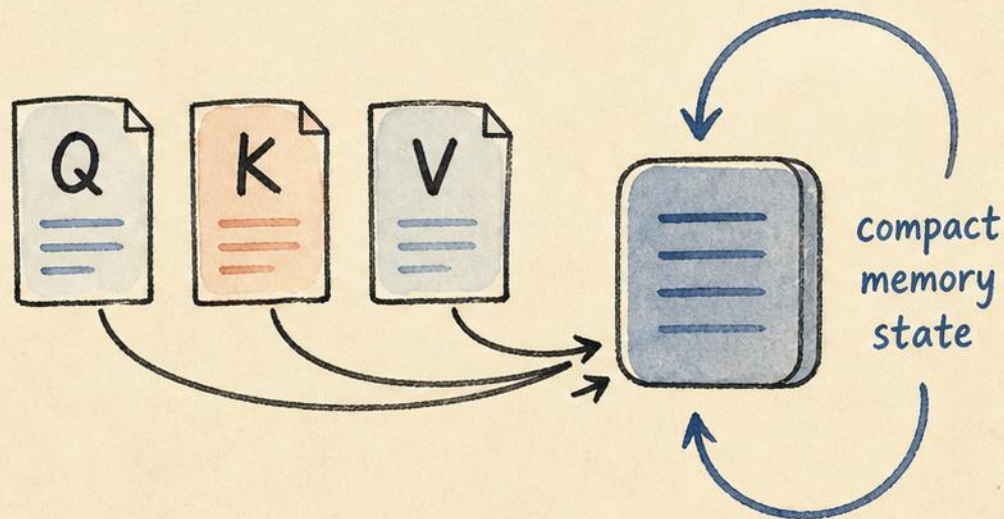


# Transformers are RNNs

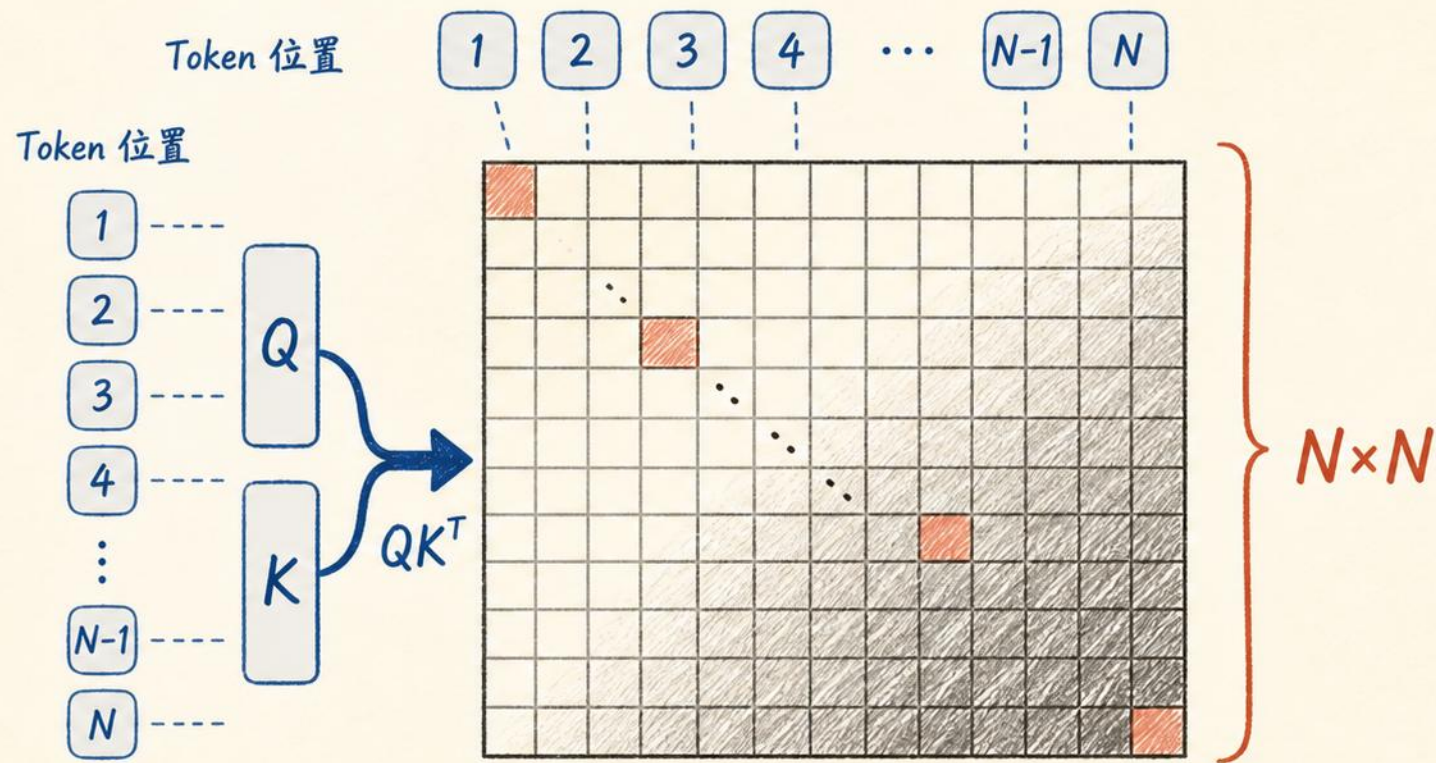
Fast Autoregressive Transformers with Linear Attention

从 QKV 到可递归状态

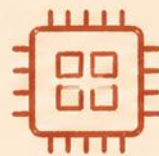
ICML 2020 · 论文阅读笔记



# 问题: 注意力矩阵随 $N^2$ 膨胀



时间  
 $O(N^2)$



显存  
 $O(N^2)$

序列越长，代价越快失控



# 关键重写：先聚合 K 与 V

标准  
Attention

$$\text{softmax}(QK^T)V$$

$QK^T: N \times N$

$$\underbrace{N \times N}_{QK^T} \times \underbrace{N \times d}_V = \underbrace{N \times d}_V$$

交换结合顺序

利用矩阵结合律

Linear  
Attention

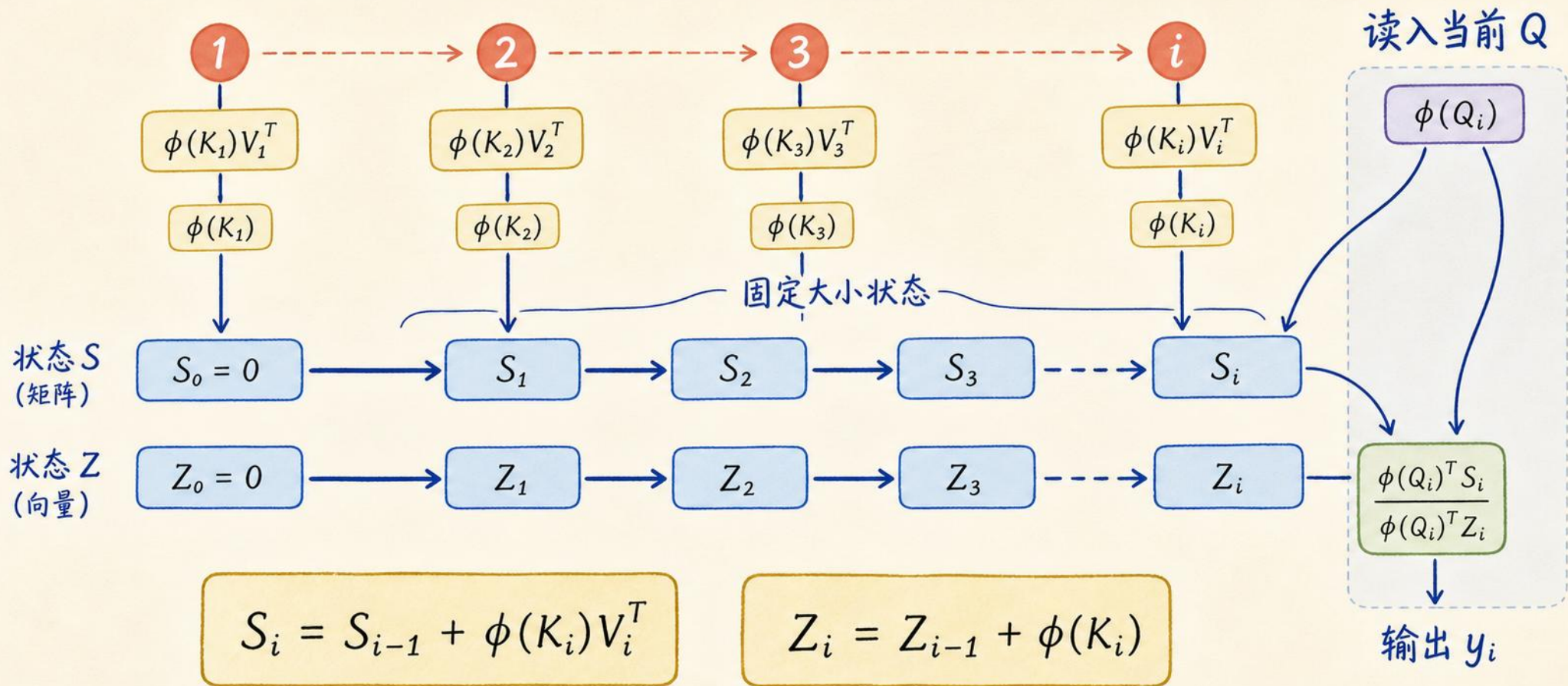
$$\phi(Q)(\phi(K)^T V)$$

$\phi(K)^T V$ : 固定状态

$$\underbrace{N \times m}_{\phi(Q)} \times \underbrace{m \times d}_{\phi(K)^T V} = \underbrace{N \times d}_V$$

# 因果机制：两个前缀状态变成 RNN

causal masking





# 证据：长序列下线性增长

- Linear:  $O(N)$
- Softmax:  $O(N^2)$
- 越长越划算

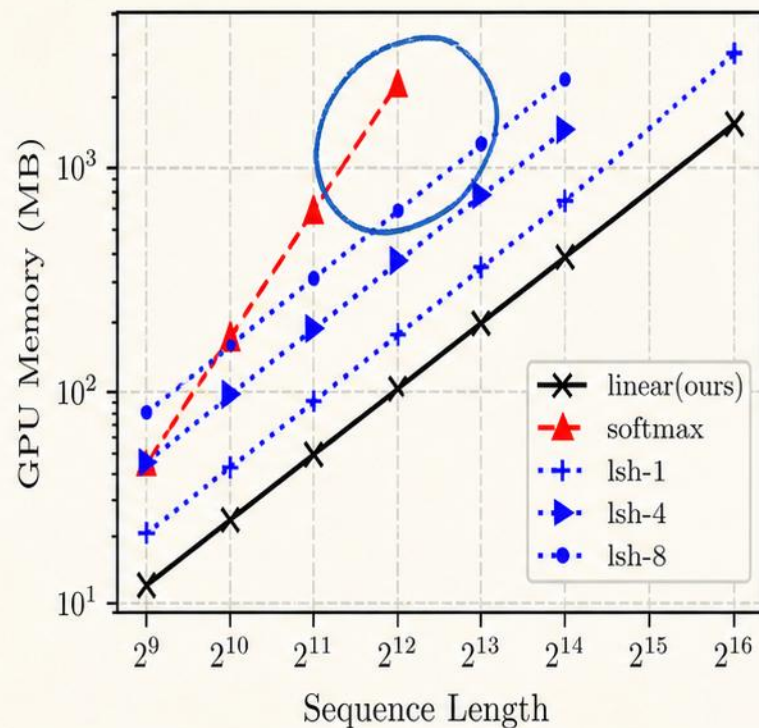
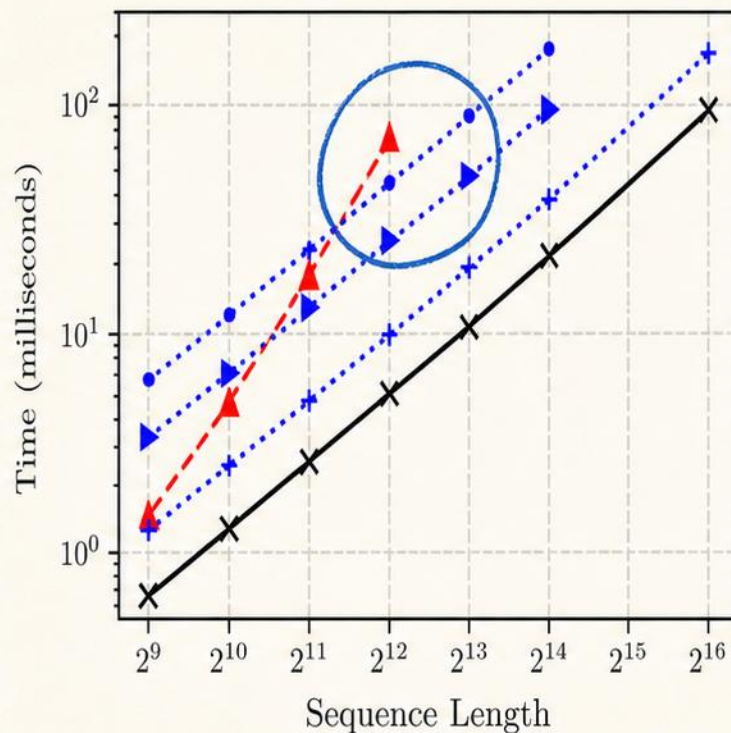
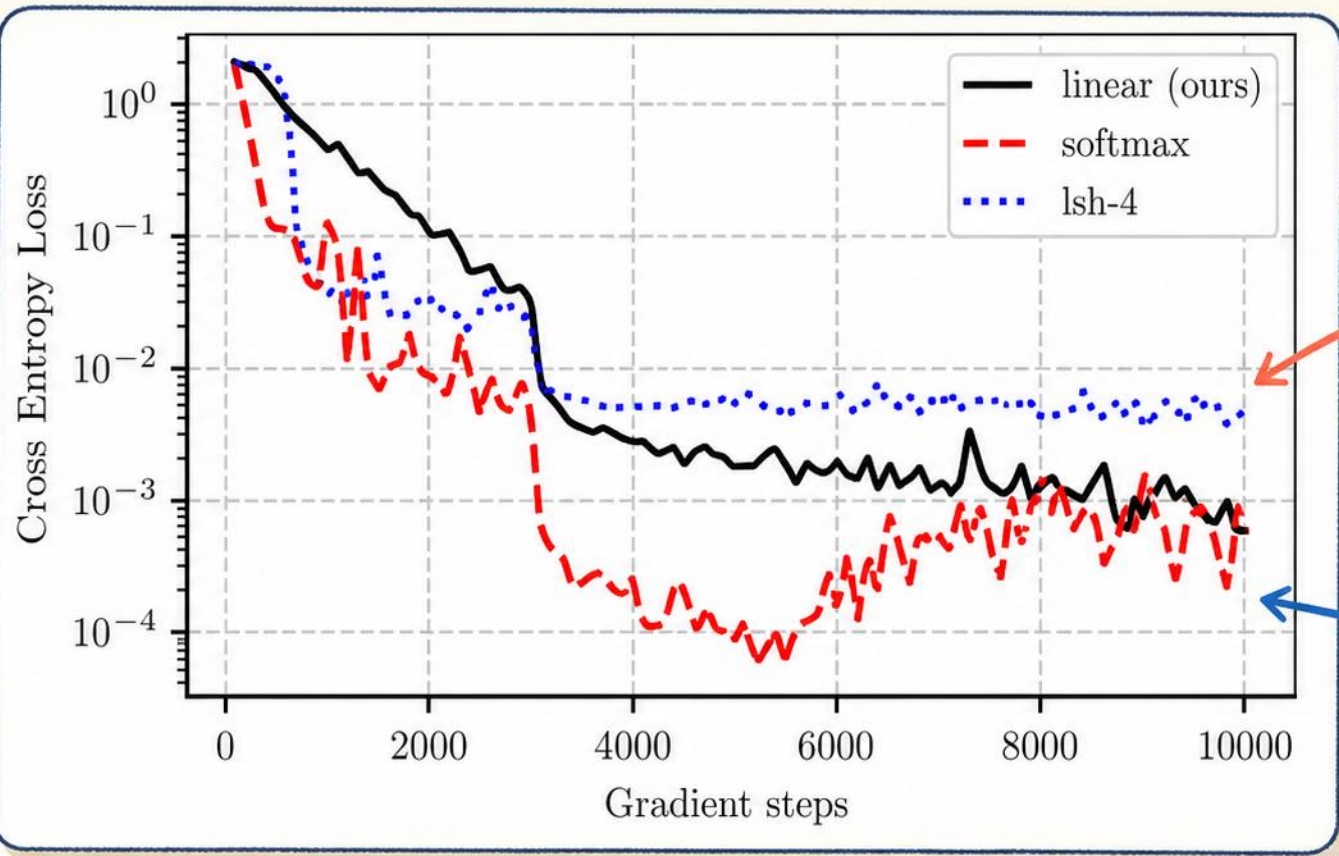


Figure 1 · synthetic benchmark

# ☆ 收敛：更快不等于训练不稳



## ☆ 稳定收敛

线性注意力 (linear) 下降平滑，  
全程稳定，无发散或震荡。

## 🎯 接近 Softmax 终点

最终交叉熵损失与 Softmax  
相当，达到相近的最优性能。

💡 copy task · causal masking



# 图像生成：长序列收益被放大

Unconditional samples



Image completion



(a)

(b)

(c)

MNIST

| Method        | Bits/dim | Images/sec          |
|---------------|----------|---------------------|
| Linear (ours) | 0.644    | <b>142.8</b> (317×) |

Unconditional samples

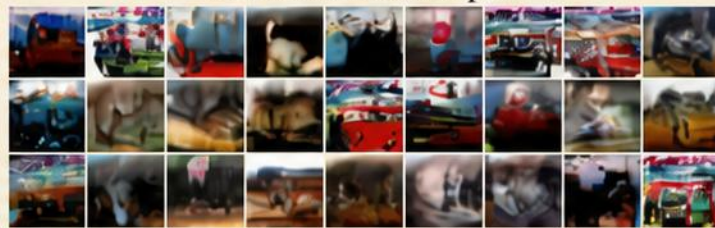
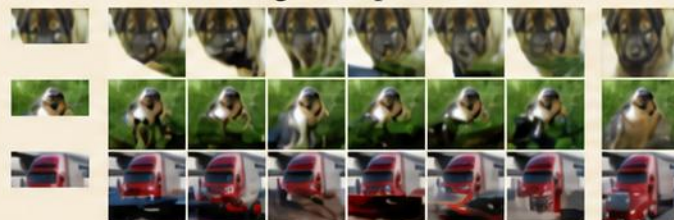


Image completion



(a)

(b)

(c)

CIFAR-10

| Method        | Bits/dim | Images/sec            |
|---------------|----------|-----------------------|
| Linear (ours) | 3.40     | <b>17.85</b> (4,462×) |

bits/dim 接近

# 语音识别：更快，但 Softmax 精度仍更高

| Method               | Validation PER | Time/epoch (s) |
|----------------------|----------------|----------------|
| Bi-LSTM              | 10.94          | 1047           |
| Softmax              | 5.12           | 2711           |
| LSH-4                | 9.33           | 2250           |
| <b>Linear (ours)</b> | <b>8.08</b>    | <b>824</b>     |

速度 ↔ 精度

Linear :  
824 s/epoch

Softmax :  
5.12 PER

- WSJ · automatic speech recognition



# 结论：什么时候值得用 Linear Attention



## 效率

- ✓ 线性复杂度
- ✓ 显著降低内存占用
- ✓ 更快的推理速度



## 表达能力

- 特征映射带来近似
- 可能损失部分建模能力
- 需根据任务权衡选择



不是无损  
Softmax

是一种可控近似，  
存在表达能力的边界

★ feature map 的选择决定取舍 ★